

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 YEARLY EXAMINATION

Mathematics Extension 1

September 2007

TIME ALLOWED: 90 minutes

Instructions:

- Write your name and class at the top of this page, and at the top of each answer sheet.
- At the end of the examination this examination paper must be attached to the front of your answers.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- **START EACH QUESTION ON A NEW PAGE**

(FOR MARKERS USE ONLY)

Q1	Q2	Q3	Q4	Q5	TOTAL
/15	/14	/14	/13	/14	/70

QUESTION 1: (15 MARKS)

Marks

1 (a) Find $\lim_{x \rightarrow \infty} \frac{2x^2 - x + 1}{3x^2 + 2}$

2 (b) If θ is an acute angle and $\sin \theta = \frac{2t}{1+t^2}$ find an expression for $\cos \theta$ in terms of t if $-1 < t < 1$

(c) Find the derivatives of the following, leaving your answer in simplest terms and including no negative indices:

2 (i) $y = \sqrt{1 - 3x^2}$

2 (ii) $y = \frac{1}{1+x^2}$

2 (iii) $y = x\sqrt{1+x}$

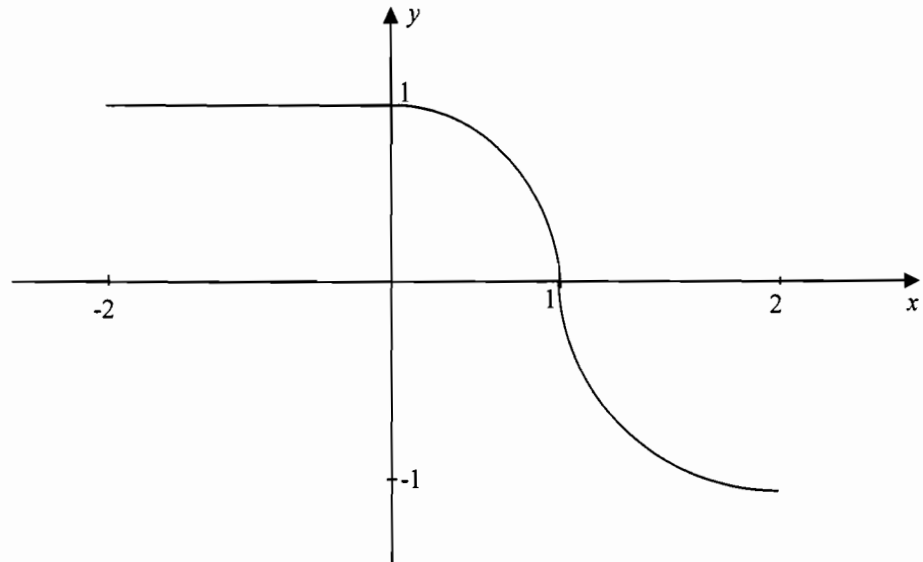
3 (d) By rationalising the numerator, find $\lim_{h \rightarrow \infty} \frac{\sqrt{2+h} - \sqrt{2}}{h}$

3 (e) Solve $\sin 2x = \cos x$ for $0^\circ \leq x \leq 360^\circ$

QUESTION 2: (14 MARKS)

Marks

- (a) The curve $y = P(x)$ is shown at below for $-2 \leq x \leq 2$



- 6 (i) Using a different set of axes for each, neatly sketch the following showing all important features

(I) $y = |P(x)|$

(II) $y = \frac{1}{P(x)}$

(III) $y = 1 - P(x)$

- 2 (ii) What is the Domain and Range of $y = \frac{1}{P(x)}$

- 2 (b) (i) Show that for all values of x , $2x^2 - 5x + 4 > 0$

- 3 (ii) For what values of x is $\frac{x^2 - 7x + 12}{2x^2 - 5x + 4} > 1$

- 1 (iii) Explain clearly why the result of part (b) (i) above is vital to the solution to part (b) (ii)

QUESTION 3: (14 MARKS)

Marks

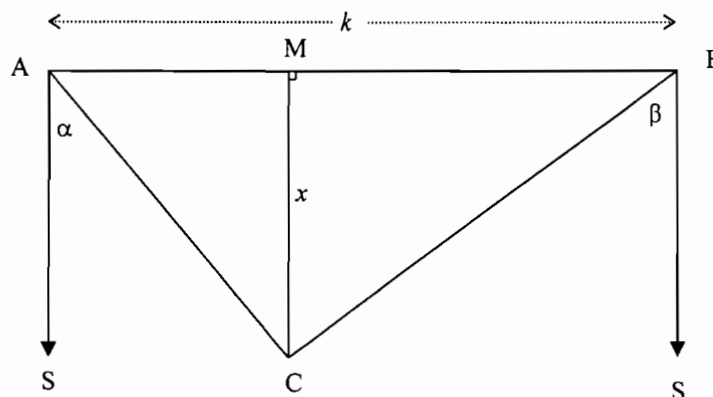
- 3 (a) The roots of the polynomial equation $2x^3 - 3x^2 + 1 = 0$ are α , β and γ
- Find (i) $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$
- (ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
- (b) Consider the interval joining A(2, -5) to B(5, 10)
- 2 (i) Find the point P which divides AB externally in the ratio 2:3
- 2 (ii) In what ratio does the point Q(4,5) divide AB?
- 3 (c) The angle between the lines $y - 2x + 5 = 0$ and $y = mx + 3$ is 45°
- Find all possible values of m .
- 3 (d) (i) Show that for the function $f(x) = \frac{2-3x}{5+x^2}$ the derivative is zero for two values of x
- 1 (ii) What does this result imply about the tangent at points with these x values?

QUESTION 4: (13 MARKS)

Marks

- 3 (a) Find the equation, in simplest expanded form, of the locus of a point $P(x,y)$ which moves so that its distance from the point $S(3,0)$ is twice its distance from the line $x-5 = 0$

- (b) A straight road runs k km from a town A due east to a town B.
A radio mast at C has a bearing of $S\alpha^\circ E$ from A and $S\beta^\circ W$ from B.
C is x km from the road AB



- (i) Copy the diagram above onto your answer sheet
- 1 (ii) Explain why $\angle ACB = \alpha + \beta$
- 2 (iii) Show that $\frac{k}{\sin(\alpha + \beta)} = \frac{BC}{\cos \alpha}$
- 2 (iv) Show that $x = \frac{k \cos \alpha \cos \beta}{\sin(\alpha + \beta)}$
- 1 (c) (i) Explain why the remainder upon dividing a polynomial $P(x)$ by $x-\alpha$ has a degree of zero
- 1 (ii) If $P(x) = 2x^4 - x^3 + 3x^2 - 1$ is divided by $x-3$ find the remainder.
- 3 (iii) One root of $ax^2 + mx + n = 0$ is three times the other.
Show that $3m^2 - 16an = 0$.

QUESTION 5: (14 MARKS)

Marks

- 2 (a) Prove that, for all values of m , the line $x = \frac{y}{m} + Am$ is a tangent to the parabola $x^2 = 4Ay$

- 3 (b) (i) Write the expression $2 \cos x - 3 \sin x$ in the form $A \cos(x + \alpha)$, showing the values of A and α

- 2 (ii) Hence, or otherwise, solve, for $0^\circ \leq x \leq 360^\circ$, the equation

$$2 \cos x - 3 \sin x = \sqrt{13}$$

Give your answer(s) to the nearest minute.

- (c) The point $P(2Ap, Ap^2)$ lies on the parabola $x^2 = 4Ay$

- 2 (i) Derive the equation of the tangent to the parabola at the point P .

- 1 (ii) The point Q is another point on the parabola and has an x -value of $\frac{2A}{p}$

Show that it has a y -value of $\frac{A}{p^2}$

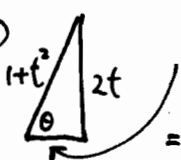
- 2 (iii) Derive the equation of the tangent to the parabola at the point Q above

- 2 (iv) The tangents at P and Q intersect at the point R .
Accurately describe the locus of R .

2007 EXT 1 MATHS Y11 FINAL EXAM

QUESTION 1

a) $\frac{2}{3}$ (1)

b)  $\sqrt{(1+t^2)^2 - 4t^2}$
 $= 1-t^2$

$\therefore \cos \theta = \frac{1-t^2}{1+t^2}$ (2)

c) i) $\frac{dy}{dx} = \frac{1}{2}(-6x)(1-3x^2)^{-\frac{1}{2}}$
 $= \frac{-3x}{\sqrt{1-3x^2}}$

ii) $\frac{dy}{dx} = -2x(1+x^2)^{-2}$
 $= \frac{-2x}{(1+x^2)^2}$

iii) $\frac{dy}{dx} = (1+x)^{\frac{1}{2}} + x \cdot \frac{1}{2}(1+x)^{-\frac{1}{2}}$
 $= \frac{2+2x+x}{2(1+x)^{\frac{1}{2}}}$
 $= \frac{2+3x}{2\sqrt{1+x}}$

d) $\frac{\sqrt{2+h} - \sqrt{2}}{h} \times \frac{\sqrt{2+h} + \sqrt{2}}{\sqrt{2+h} + \sqrt{2}}$
 $= \frac{2+h-2}{h(\sqrt{2+h} + \sqrt{2})}$
 $= \frac{1}{\sqrt{2+h} + \sqrt{2}}$

$\therefore \lim_{h \rightarrow \infty} \left(\frac{\sqrt{2+h} - \sqrt{2}}{h} \right) = 0$

e) $2\sin x \cos x = \cos x$

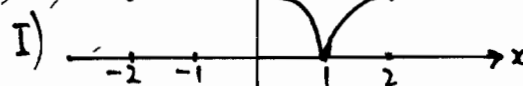
ie. $\cos x(2\sin x - 1) = 0$

$\therefore \cos x = 0$ OR $\sin x = \frac{1}{2}$

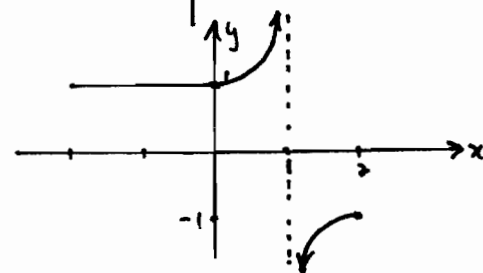
$\therefore x = 90^\circ, 270^\circ$ $\therefore x = 30^\circ, 150^\circ$

QUESTION 2

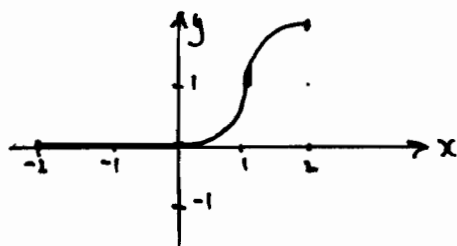
a) i)



II)



III)



ii) $\mathcal{D}: \{x: -2 \leq x \leq 2, x \neq 1\}$

OR $\mathcal{D}: \{x: -2 \leq x < 1\} \cup \{x: 1 < x \leq 2\}$

$\mathcal{R}: \{y: -\infty < y \leq -1\} \cup \{y: 1 \leq y < \infty\}$

b) i) For $2x^2 - 5x + 4$,

$\Delta = 25 - 4 \cdot 2 \cdot 4$

$= -7$

Since $\Delta < 0$, $2x^2 - 5x + 4$ is positive definite, ie. > 0

OR $2x^2 - 5x + 4 = 2(x - \frac{5}{4})^2 + \frac{7}{8}$

which has a minimum value of $\frac{7}{8}$

$\therefore 2x^2 - 5x + 4 > 0$

$\therefore x < \frac{1}{2} \text{ or } x > \frac{5}{2}$

QUESTION 2 (cont)

b) ii) $\frac{x^2 - 7x + 12}{2x^2 - 5x + 4} > 1$

ie. $x^2 - 7x + 12 > 2x^2 - 5x + 4$ — (A)

ie. $x^2 + 2x - 8 < 0$

$(x+4)(x-2) < 0$

$\therefore -4 < x < 2$

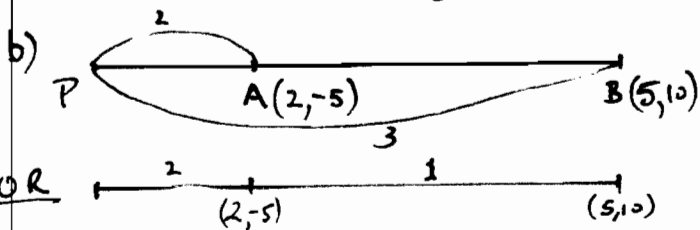
iii) Inequality sign at (A) above does not have to be reversed because we know we've multiplied both sides by a positive quantity.



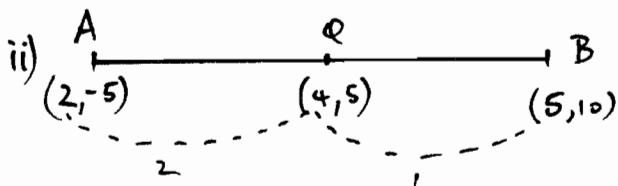
QUESTION 3

a) i) $\alpha + \beta + \gamma = \frac{3}{2}, \left(-\frac{b}{a}\right)$
 $\alpha\beta\gamma = -\frac{1}{2}, \left(-\frac{d}{a}\right)$

ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$
 $= \frac{0}{-\frac{1}{2}}$
 $= 0$



i) At P $x = -4, y = -35$



Q divides AB in the ratio 2:1.

(For real numbers, better not

c) $m_1 = m, m_2 = 2$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

ie $1 = \left| \frac{m - 2}{1 + 2m} \right|$

ie $m - 2 = 1 + 2m \Rightarrow m = -3$

OR $2 - m = 1 + 2m \Rightarrow m = \frac{1}{3}$

d) i)

$f'(x) = \frac{(5+x^2)(-3) - (2-3x)(2x)}{(5+x^2)^2}$
 $= \frac{3x^2 - 4x - 15}{(5+x^2)^2}$

$= 0$ when $3x^2 - 4x - 15 = 0$

ie. $(3x+5)(x-3) = 0$

$\therefore x = -\frac{5}{3}, 3$

ii) When $x = -\frac{5}{3}$ or 3 the tangent is horizontal ($\parallel$ the x axis)

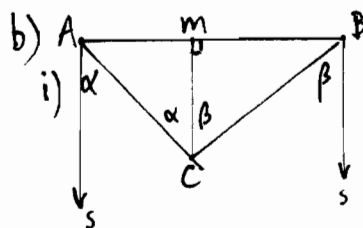
QUESTION 4

a)

$PS = 2PM$
 $PS^2 = 4PM^2$
 $(x-3)^2 + (y-0)^2 = 4(5-x)^2$

$x^2 - 6x + 9 + y^2 = 4(25 - 10x + x^2)$

ie $3x^2 - 34x + 91 - y^2 = 0$



ii) $\angle ACB = \angle ACM + \angle BCM$

But $\angle ACM = \alpha$ (alternate $\angle$ s, parallel lines)

QUESTION 4 (cont.)

b) iii) In $\triangle ABC$, $\frac{AB}{\sin \hat{ACB}} = \frac{BC}{\sin \hat{CAB}}$

ie. $\frac{k}{\sin(\alpha+\beta)} = \frac{BC}{\sin(90-\alpha)}$

But $\sin(90-\alpha) = \cos \alpha$

$\therefore \frac{k}{\sin(\alpha+\beta)} = \frac{BC}{\cos \alpha}$

iv) In $\triangle BMC$, $\frac{x}{BC} = \cos \beta$

ie. $x = BC \cos \beta$

From iii) above, $BC = \frac{k \cos \alpha}{\sin(\alpha+\beta)}$

$\therefore x = \frac{k \cos \alpha \cos \beta}{\sin(\alpha+\beta)}$ as required

c) i) Remainder must have a degree less than the divisor, $(x-\alpha)$, which is of degree one.

ii) Remainder = $P(3)$
 $= 2 \times (3)^4 - 3^3 + 3(3)^2 - 1$
 $= 161$

iii) Let the roots of $ax^2 + mx + n = 0$ be α and 3α .

Then $4\alpha = -\frac{m}{a}$ and $3\alpha^2 = \frac{n}{a}$
 $\therefore \alpha = -\frac{m}{4a}$ and $3\left(-\frac{m}{4a}\right)^2 = \frac{n}{a}$
 ie. $3m^2 = 16an$

QUESTION 5

(a) $x = \frac{y}{m} + Am$ and $x^2 = 4Ay$

intersect when $\left(\frac{y}{m} + Am\right)^2 = 4Ay$.

ie. $\frac{y^2}{m^2} + 2Ay + A^2m^2 - 4Ay = 0$

ie. $y^2 - 2Aym^2 + A^2m^4 = 0$

Now $\Delta = 4A^2m^4 - 4 \cdot A^2m^4 = 0$

$\therefore$ Only 1 point of intersection

$\therefore x = \frac{y}{m} + Am$ is a tangent to the parabola.

(b) i) $2 \cos x - 3 \sin x = A \cos(x+\alpha)$

$\therefore A = \sqrt{2^2 + 3^2} = \sqrt{13}$

$\therefore \sqrt{13} \cos(x+\alpha) = 2 \cos x - 3 \sin x$

From this we get

$\cos \alpha = \frac{2}{\sqrt{13}}$ and $\sin \alpha = \frac{3}{\sqrt{13}}$

$\therefore \alpha = 56^\circ 19'$

$\therefore 2 \cos x - 3 \sin x = \sqrt{13} \cos(x + 56^\circ 19')$

ii) $2 \cos x - 3 \sin x = \sqrt{13}$

$\therefore \cos(x + 56^\circ 19') = 1$

ie $x + 56^\circ 19' = 0^\circ$ or 360°

$\therefore x = -56^\circ 19'$ or $303^\circ 41'$

But $-56^\circ 19'$ is outside the domain

$\therefore x = 303^\circ 41'$

(c) i) $y = \frac{x^2}{4A}$

$\therefore \frac{dy}{dx} = \frac{2x}{4A} = \frac{x}{2A}$

At $P(2Ap, Ap^2)$ $m_T = \frac{2Ap}{2A} = p$

and the equation of the tangent is $y - Ap^2 = p(x - 2Ap)$

ie. $y = px - Ap^2$

QUESTION 5 (cont.)

c) ii) $x^2 = 4Ay$ and $x = \frac{2A}{p}$

$$\therefore \frac{4A^2}{p^2} = 4Ay$$

$$\text{ie } y = \frac{4A^2}{4Ap^2} = \frac{A}{p^2}$$

$$\text{iii) At } Q, m_T = \frac{2A/p}{2A} \\ = \frac{1}{p}$$

$\therefore$ Equation of tangent

$$\text{is } y - \frac{A}{p^2} = \frac{1}{p} \left(x - \frac{2A}{p} \right)$$

$$y - \frac{A}{p^2} = \frac{x}{p} - \frac{2A}{p^2}$$

$$\text{ie. } p^2 y - A = px - 2A$$

$$\text{ie. } p^2 y = px - A$$

or

$$y = \frac{x}{p} - \frac{A}{p^2}$$

$$\text{iv) Now } y = px - Ap^2 \\ \text{and } p^2 y = px - A$$

$$\therefore y(1 - p^2) = -A(p^2 - 1)$$

$$\therefore y = A$$

The locus of R
lies on the extensions of
the latus rectum.